

Cosmic Filament Detection Pipeline — Technical Summary (V1–V10)

This document summarizes the development of a custom computational pipeline designed to detect and analyze filamentary structures in the large-scale structure of the universe (cosmic web). The pipeline was implemented in Python and executed on Google Colab using public SDSS galaxy catalog data. The goal is to automatically identify statistically significant filamentary structures within 3D galaxy distributions.

Dataset: Sloan Digital Sky Survey (SDSS) galaxy sample.

Approximate size used in testing: 50,000 galaxies.

Input coordinates: Right Ascension (RA), Declination (DEC), Redshift (z).

Coordinates converted to normalized 3D Cartesian positions for spatial analysis.

Pipeline Evolution

Version	Main Function
V1–V3	Initial filament detection and backbone extraction algorithms
V4	Backbone purification and improved straightness metrics
V5	Monte Carlo significance test for filament detection
V6	Global cosmic web density mapping and graph visualization
V7	Local filament hunter scanning dense regions
V8	Filament catalog builder and ranking of candidate structures
V9	Graph spine extraction to produce a clean filament spine
V10	Multi-seed robustness testing to measure algorithm stability

Current Results

Key results obtained from the current pipeline stage:

- Filament structures can be extracted from dense cosmic web regions.
- Statistical significance tests show extremely high deviation from random distributions (z-score > 16 in some tests).
- Multi-seed robustness testing confirms that filamentary structures consistently appear, although branch selection may vary.
- Straightness metrics allow ranking and comparison between filament candidates.

Robustness Test (V10)

Robustness summary (V10 test): Mean straightness ≈ 0.39 Standard deviation ≈ 0.16 Mean filament length ≈ 0.24 (normalized coordinate units) Robustness score ≈ 0.86

Next Development Steps

Planned improvements for next versions: • Branch selection algorithm to identify the most linear filament path • Automatic ranking of multiple filament candidates • Performance scaling for larger galaxy catalogs (millions of objects) • Production of publication-quality visualizations and exportable catalogs